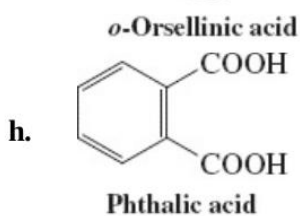
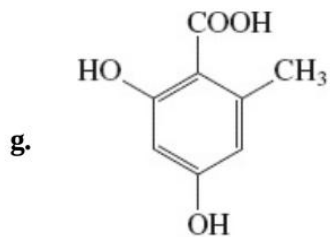
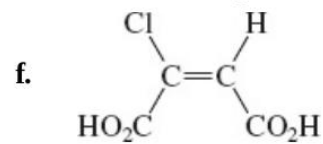
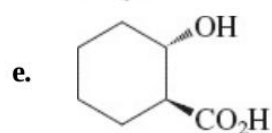
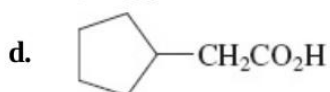
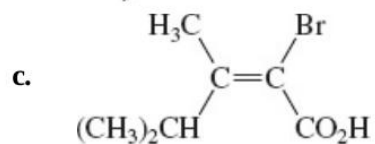
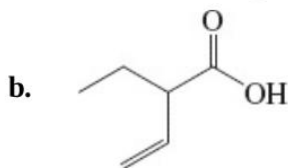
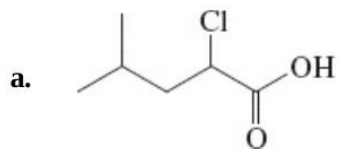
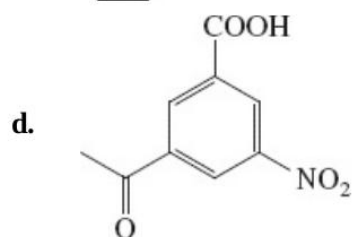
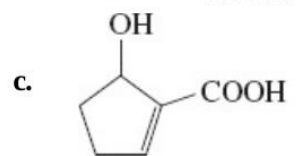
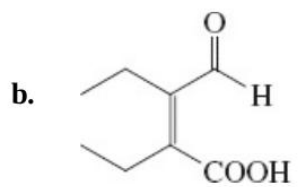
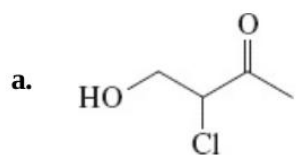


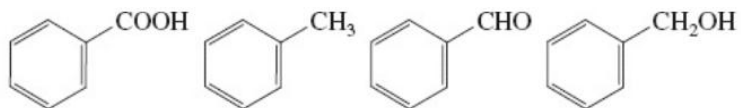
27. Name (IUPAC or *Chemical Abstracts* system) or draw the structure of each of the following compounds.



28. Name each of the following compounds according to IUPAC or *Chemical Abstracts*. Pay attention to the choice of the main chain and the order of precedence of the functional groups.



29. Rank the group of molecules shown here in decreasing order of boiling point and solubility in water. Explain your answers.

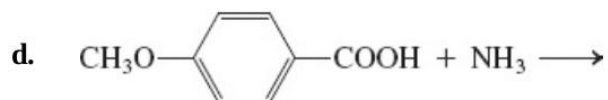
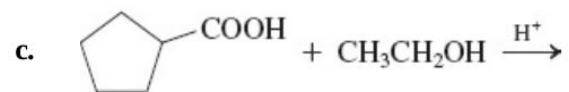
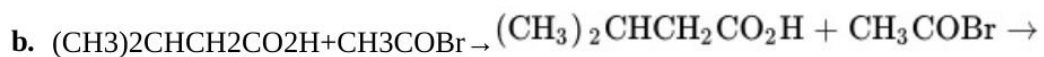
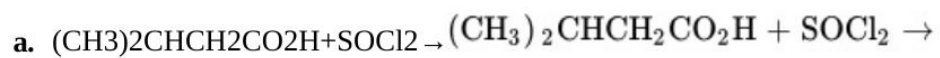


- 30.** Rank each of the following groups of organic compounds in order of decreasing acidity.

31. Propose a structure for a compound that displays the spectroscopic data that follow. The molecular ion in the mass spectrum appears at $m/z=116$. IR: $\tilde{\nu}=1710$ (s), 3000 (s, broad) cm^{-1} ; $^1\text{H NMR}$: $\delta=0.94$ (t, $J=7.0$ Hz, 6 H), 1.59 (m, 4 H), 2.36 (quin, $J=7.0$ Hz, 1 H), 12.04 (broad s, 1 H) ppm; $^{13}\text{C NMR}$: $\delta=11.7, 24.7, 48.7, 183.0$ ppm.

32. **(a)** An unknown A has the formula $C_7H_{12}O_2$ and infrared spectrum A (below). To which class does this compound belong? **(b)** Use the other spectra (NMR-B, p. 961, and F, p. 962; IR-D, E, and F, pp. 961-962) and spectroscopic and chemical information in the reaction sequence to determine the structures of A and the other unknown substances B through F. References are made to relevant sections of earlier chapters, but do not look them up before you have tried to solve the problem without the extra help. **(c)** Another unknown compound, G,

33. Give the products of each of the following reactions.



e. Product of (d), heated strongly

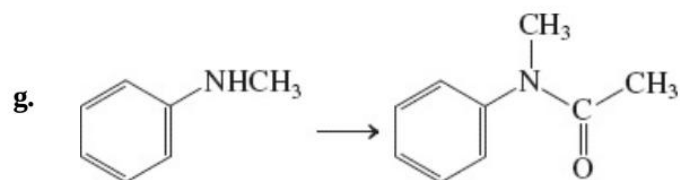
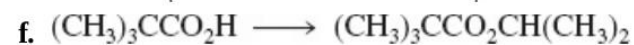
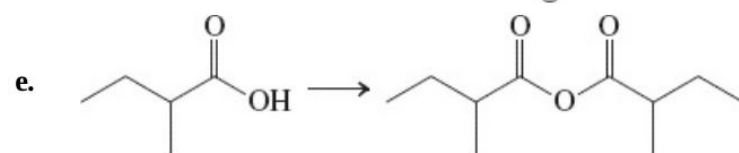
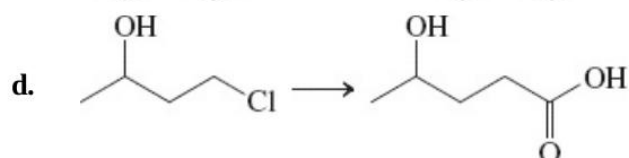
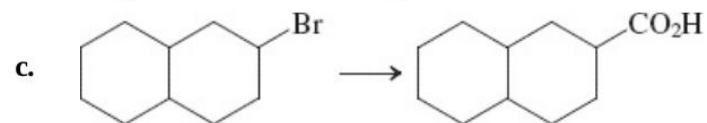
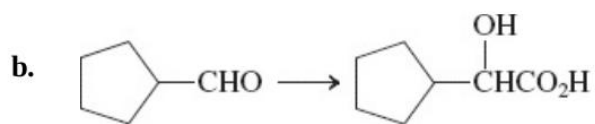
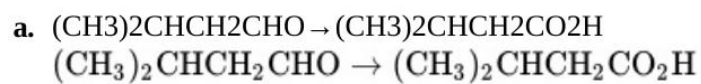
f. Phthalic acid [[Problem 27](#), part (h)], heated strongly

34. When 1,4- and 1,5-dicarboxylic acids, such as butanedioic (succinic) acid ([Section 19-8](#)), are treated with SOCl_2 or PBr_3 in attempted preparations of the diacyl halides, the corresponding cyclic anhydrides are obtained. Explain mechanistically.

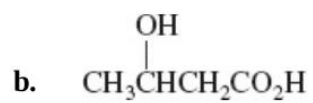
35. Reaction review. Without consulting the Reaction Road Map on [p. 958](#), suggest reagents to convert each of the following starting materials into hexanoic acid: **(a)** hexanal; **(b)** methyl hexanoate; **(c)** 1-bromopentane; **(d)** 1-hexanol; **(e)** hexanenitrile.

36. Reaction review II. Without consulting the Reaction Road Map on [p.](#)

37. Fill in suitable reagents to carry out the following transformations.

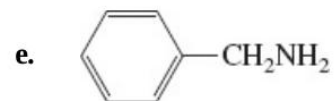
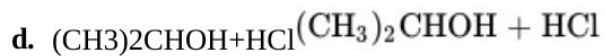


38. Propose syntheses of each of the following carboxylic acids that employ at least one reaction that forms a carbon-carbon bond.

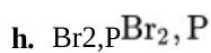
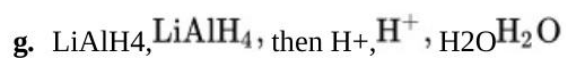


39. (a) Write a mechanism for the esterification of propanoic acid with ^{18}O -labeled ethanol. Show clearly the fate of the ^{18}O label. (b) Acid-catalyzed hydrolysis of an unlabeled ester with ^{18}O -labeled water (H_2^{18}O) leads to incorporation of some ^{18}O into *both* oxygens of the carboxylic acid product. Explain by a mechanism. (**Hint:** You must use the fact that all steps in the mechanism are reversible.)

40. Give the products of reaction of propanoic acid with each of the following reagents.



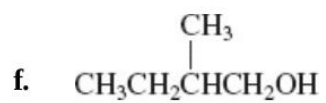
f. Product of (e), heated strongly



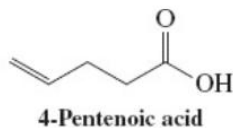
- 41.** Give the product of reaction of cyclopentanecarboxylic acid with each of the reagents in [Problem 40](#).

44. Suggest a preparation of hexanoic acid from pentanoic acid.

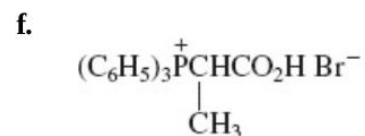
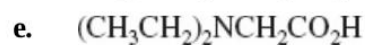
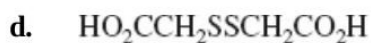
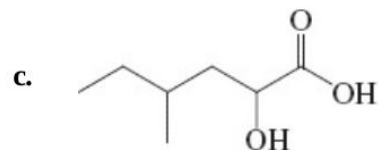
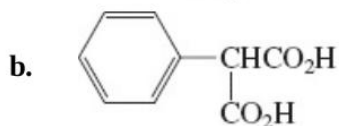
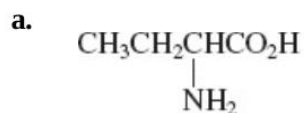
45. Give reagents and reaction conditions that would allow efficient conversion of 2-methylbutanoic acid into **(a)** the corresponding acyl chloride; **(b)** the corresponding methyl ester; **(c)** the corresponding ester with 2-butanol; **(d)** the anhydride; **(e)** the *NN*-methanide;



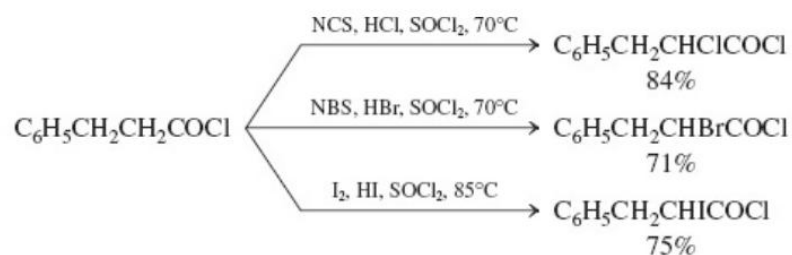
46. Treatment of 4-pentenoic acid (below) with Br_2 in the presence of dilute aqueous base yields a nonacidic material with the formula $\text{C}_5\text{H}_7\text{BrO}_2$. **(a)** Suggest a structure for this compound and a mechanism for its formation. **(b)** Can you find a second, isomeric product whose formation is also mechanistically reasonable? **(c)** Discuss the issues that contribute to determining which of the two is the major product in this reaction. (**Hint:** Review [Section 12-6](#).)



47. Show how the Hell-Volhard-Zelinsky reaction might be used in the synthesis of each of the following compounds, beginning in each case with a simple monocarboxylic acid. Write detailed mechanisms for all the reactions in *one* of your syntheses.

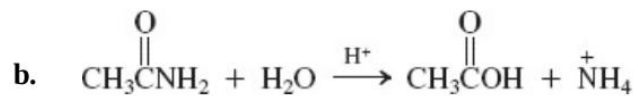
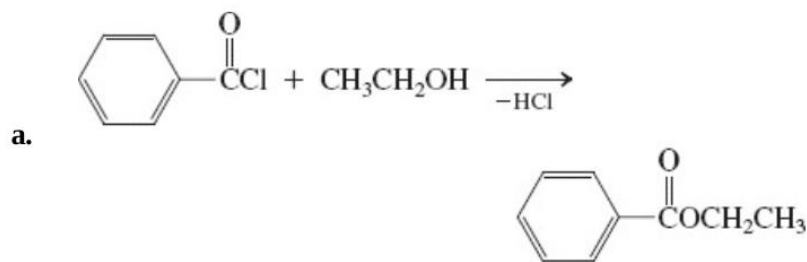


48. The Hell-Volhard-Zelinsky reaction produces only bromocarboxylic acids. However, modified procedures have been developed to convert acyl chlorides into $\alpha\alpha$ -chloro- and $\alpha\alpha$ -bromoacyl chlorides by reaction with *NN*-chloro- and *NN*-bromobutanamide (*NN*-chloro- and *NN*-bromosuccinimide, NCS and NBS: [Section 14-2](#)), respectively. Reaction with I_2 gives $\alpha\alpha$ -iodoacyl chlorides. (SOCl_2 is used as the solvent to maintain the acyl chloride functional group.) Suggest a mechanism for any one of these processes.



50. Attempted $\text{CrO}_3/\text{CrO}_3$ oxidation of 1,4-butanediol to butanedioic acid results in significant yields of “ γ - γ -butyrolactone.” Explain mechanistically.

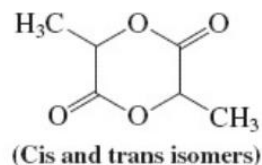
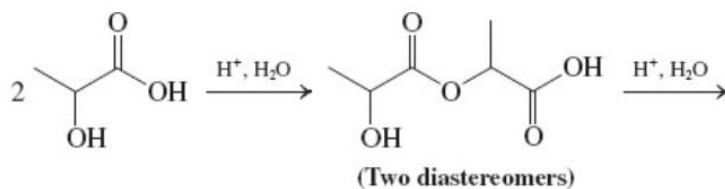
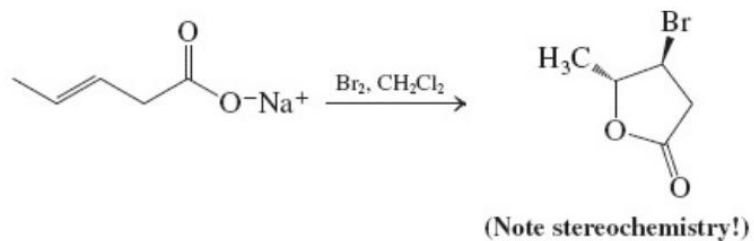
51. Following the general mechanistic scheme given in [Section 19-7](#), write detailed mechanisms for each of the following substitution reactions. (Note: These transformations are part of [Chapter 20](#), but try to solve the problem without looking ahead.)



52. Suggest structures for the products of each reaction in the following synthetic sequence.

56. Propose a short synthesis of 2-butyne-1-carboxylic acid, $\text{CH}_3\text{C}\equiv\text{CCO}_2\text{H}$, starting from propyne. (**Hint:** Review [Sections 13-2](#) and [13-5](#).)

58. [Section 19-9](#) showed you that 4- and 5-hydroxy acids can undergo acid-catalyzed intramolecular esterification to produce the corresponding lactone in good yield. Consider the following two examples of lactonization. Divide the analysis of the reaction sequences among your group. Propose reasonable mechanisms to explain the respective product formation.



Reconvene to present your mechanisms to each other.